# Intelligent News Credibility Analysis

Project Number: Project 11

Milestone 1 – Traditional Machine Learning & NLP

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**Team Size: 4 Members**

# Abstract

This report presents the successful completion of Milestone 1 for the Intelligent News Credibility Analysis project. The primary objective of this phase was to design, implement, and rigorously evaluate a robust fake news detection system using traditional Machine Learning (ML) and Natural Language Processing (NLP) techniques. In accordance with project constraints, this milestone strictly excludes the use of Generative AI, Large Language Models (LLMs), or Agentic AI frameworks, focusing instead on interpretable, computationally efficient, and well-validated classical approaches.

The system was developed using the ISOT Fake News Dataset, which contains labeled news articles categorized as fake or real. A structured and modular pipeline was implemented, beginning with comprehensive text preprocessing, including normalization, stop-word removal with negation preservation, lemmatization, duplicate elimination, and noise reduction. The cleaned textual data was transformed into numerical feature representations using TF-IDF vectorization with unigram and bigram configurations, enabling effective capture of contextual relationships while controlling dimensionality.

To ensure rigorous model selection, multiple baseline classifiers were evaluated, including Logistic Regression, Multinomial Naive Bayes, and Linear Support Vector Machine (SVM). Performance comparison was conducted using F1-score as the primary evaluation metric to balance precision and recall. Following comparative analysis, Linear SVM demonstrated superior performance on high-dimensional sparse feature space. Hyperparameter optimization was performed using 3-fold cross-validation to fine-tune the regularization parameter, ensuring generalization stability and minimizing overfitting risk.

The final optimized model achieved approximately 99 percent accuracy and F1-score on unseen test data, demonstrating strong predictive capability and balanced classification across both fake and real news categories. The resulting system is modular, deployment-ready, and scalable, providing a solid analytical foundation for subsequent user interface integration and public deployment in future project milestones.

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# 1 Introduction

The rapid spread of misinformation across digital platforms has made fake news detection a significant societal challenge. Manual verification is time-consuming and reactive. Machine learning enables scalable and proactive credibility assessment.

Milestone 1 focuses on building an interpretable and well-validated binary classification system (Fake vs Real news) using classical ML techniques.

# 2 Dataset Description

The ISOT Fake News Dataset consists of two labeled files: Fake.csv and True.csv.

- The `title` and `text` columns were merged to preserve contextual information.
- Duplicate entries were removed to prevent data leakage.
- Missing or empty records were filtered.
- Data was shuffled and split into 80% training and 20% testing sets.

# 3 Methodology

## 3.1 Text Preprocessing

The preprocessing pipeline includes:

- Lowercasing
- Removal of URLs, numbers, and punctuation
- Stop-word removal while preserving negations
- Lemmatization
- Duplicate elimination

Preserving negation words prevents semantic distortion in factual statements.

### 3.2 Feature Engineering

TF-IDF vectorization was configured as:

- ngram_range = (1,2)
- max_features = 15000
- min_df = 5
- max_df = 0.9

This configuration balances contextual richness and dimensionality control.

## 4 Design Justification

### 4.1 Choice of Traditional Machine Learning

Traditional ML models are computationally efficient and well-suited for high-dimensional sparse data. They align with Milestone 1 constraints that exclude Generative AI.

### 4.2 Choice of TF-IDF Representation

TF-IDF was chosen over word embeddings because it preserves interpretability, reduces computational complexity, and performs effectively in linearly separable text classification tasks.

### 4.3 Choice of Linear SVM

Linear SVM achieved the highest F1-score among evaluated models. It is particularly effective in sparse high-dimensional feature spaces and maximizes margin separation between classes.

### 4.4 Evaluation Strategy

F1-score was selected to balance precision and recall. GridSearchCV with 3-fold cross-validation ensured model stability and reduced overfitting risk.

# 5 Model Evaluation

## 5.1 Baseline Comparison

| Model | F1 Score | Assessment |
|---|---|---|
| Linear SVM | 0.9942 | Best Performing |
| Logistic Regression | 0.9876 | Strong Baseline |
| Naive Bayes | 0.9530 | Acceptable Baseline |

Table 1: Baseline Model Comparison

## 5.2 Final Performance

| Class | Precision | Recall | F1-score |
|---|---|---|---|
| Fake (0) | 0.99 | 0.99 | 0.99 |
| Real (1) | 0.99 | 1.00 | 0.99 |

Table 2: Final Classification Report

## 5.3 Confusion Matrix

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

| Actual / Predicted | Fake | Real |
|---|---|---|
| Fake | 4640 | 56 |
| Real | 32 | 4210 |

Table 3: Confusion Matrix

The low false positive and false negative values demonstrate balanced and reliable classification performance across both classes.

# 6 System Architecture

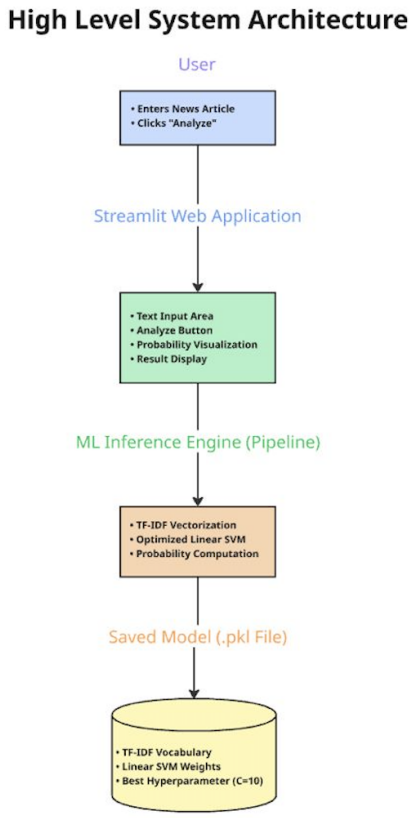


Figure 1: High-Level System Architecture

The system follows a layered architecture that separates user interaction, application logic, and machine learning inference to ensure modularity and scalability. The user interacts with the Streamlit web application by entering a news article. The web interface forwards the input text to the ML inference engine, which consists of the saved trained pipeline (TF-IDF vectorizer + optimized Linear SVM model).

The inference engine preprocesses the text, transforms it into TF-IDF feature vectors, and applies the optimized classifier to generate predictions. The system outputs the predicted class (Fake/Real) along with a confidence score, along with text analysis including word count, Character Count and Unique words, also the top 10 most frequent words precedency. The trained model is stored as a serialized `.pkl` file, containing the TF-IDF vocabulary, learned model weights, and optimal hyperparameters. This separation of layers improves maintainability, deployment flexibility, and scalability.

# 7 Machine Learning Pipeline

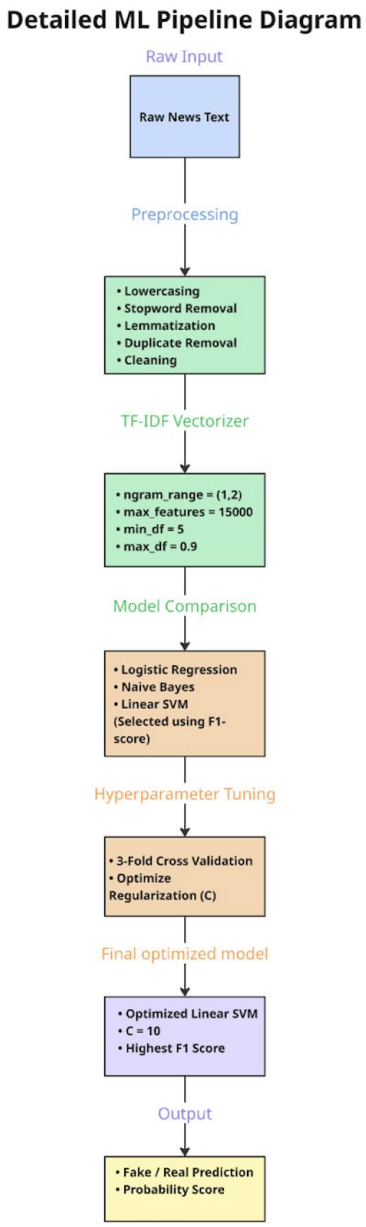


Figure 2: Training and Inference Workflow

The ML pipeline transforms raw news text through preprocessing (cleaning, stop-word removal with negation preservation, and lemmatization) followed by TF-IDF vectorization using unigrams and bigrams. Multiple models were evaluated, and Linear SVM was selected based on the highest F1-score. Hyperparameter tuning using 3-fold cross-validation optimized the regularization parameter, resulting in the final deployed model.

# 8 Contribution Matrix

| Task | Khushi | Manpreet | Avneet | Riya |
|---|---|---|---|---|
| Dataset Preparation | Lead | | Support | |
| Preprocessing Pipeline | Lead | | | |
| Feature Engineering | | Lead | | Support |
| Model Training | Support | Lead | | |
| Hyperparameter Tuning | | Lead | Support | |
| UI Development | | | Lead | |
| Documentation | | | | Lead |
| Repository Management | | | | Lead |

Table 4: Team Contribution Matrix

# 9 Limitations

The model may reflect dataset-specific vocabulary patterns and requires validation on external datasets. Traditional ML approaches lack deeper semantic reasoning compared to transformer-based models.

# 10 Future Work

- External dataset validation
- Explainability integration (SHAP/LIME)
- Transformer benchmarking
- Deployment optimization

# 11 Conclusion

Milestone 1 successfully established a modular and high-performing fake news classification system. The optimized Linear SVM achieved approximately 99% F1-score on unseen test data. The results confirm that classical machine learning methods, when combined with effective feature engineering and rigorous validation, remain highly competitive for text classification tasks. The developed system provides a strong and scalable foundation for subsequent project milestones.